

AI-Based IDS/IPS PortScan & Network Reconnaissance

Decision-Making Needs Expression

AI Lifecycle | Full CRISP-DM Scope

Module:	Artificial Intelligence
Attack Focus:	Port Scanning & Network Reconnaissance
Status:	Academic Project Final Submission
Team:	5 Members (ENSA Engineering)
Version:	Hybrid BRD v2.0 (April 2026)
Severity:	MEDIUM-HIGH

Hybrid Design: BRD AI Core + Honeypot Response Layer
+ 5-Person Team Roles + Week-by-Week Execution Timeline

1. Executive Summary

This document presents the formal Needs Expression for the design, implementation, testing, and validation of an AI model dedicated to real-time detection and automated mitigation of Port Scanning and Network Reconnaissance attacks within a controlled LAN.

Reconnaissance is the first phase of any cyberattack kill chain. Attackers probe the network to map active hosts, open ports, and running services before launching a targeted attack. Early detection and containment of this phase prevents subsequent, more destructive intrusions.

This Hybrid BRD v2.0 extends the original scope with: (1) an optional Honeypot redirect layer for attacker intelligence gathering, (2) a week-by-week execution timeline, and (3) explicit 5-person team role assignments all aligned to the core AI/ML pipeline.

2. Problem Statement

Port scanning in its various forms (SYN, ACK, UDP, XMAS, FIN scan) produces distinctive traffic patterns fundamentally different from normal user activity. An attacker running Nmap sends probes to hundreds of ports in rapid succession, producing anomalous SYN packet distributions, RST responses, and unanswered connections.

The core challenge is detecting stealthy or slow-rate scans where attackers deliberately space probes over time to evade threshold-based IDS systems. This requires ML models capable of identifying behavioral anomalies across time windows, not instantaneous packet analysis.

3. Project Objectives

Objective	Description
Primary Detection	Train ML model to detect port scanning and recon traffic on the LAN
Scan Type Coverage	Detect SYN Scan (stealth), TCP Connect, UDP, ACK, and XMAS variants
Slow Scan Detection	Identify slow/distributed scans that evade threshold-based IDS
Attacker Profiling	Profile: source IP, MAC, scan type inferred, ports probed, duration
Real-time Alerting	Trigger alerts with scan type, confidence score, and risk level
Automated Response	Block scanning IP via iptables; optional honeypot redirect

4. Functional Requirements

4.1 System Inputs

- * Raw TCP/UDP packets on LAN interface, filtered for SYN-only or partial handshake connections
- * Training datasets: CICIDS2017 (PortScan ~158K flows) and UNSW-NB15 (Reconnaissance ~13K records)
- * Sliding time-window aggregated features per source IP: 10-second and 60-second windows
- * Network metadata: src/dst IP, MAC, port, protocol, TCP flags, TTL, timestamp

4.2 System Outputs

- * Real-time dashboard: unique ports probed per source IP, port heat map, active scanners list
- * Classification per window: BENIGN or PORT_SCAN with inferred scan type and confidence score
- * Attacker profile: source IP, MAC, targeted hosts, port range probed, first/last seen timestamps
- * Alert with scan type label (SYN / Full Connect / UDP / Stealth) and severity score
- * iptables DROP rule for scanning source IP with event logged to database
- * [OPTIONAL] Honeypot: redirect scanner to fake listener on original IP, log all probes

5. Dataset Specification

Dataset	Subset	Samples (approx.)	Source	Status
CICIDS2017	PortScan category	~158,000 flows	UNB / CIC	PRIMARY
UNSW-NB15	Reconnaissance category	~13,000 records	UNSW Australia	SECONDARY
CAIDA Telescope	Backscatter / SYN probes	Variable	CAIDA.org	OPTIONAL

Dataset Rationale

CICIDS2017: modern IDS benchmark with labeled attack flows and realistic traffic patterns.

UNSW-NB15: adds diversity different capture environment and attack representation.

KDD-99 (1999) explicitly REJECTED universally obsolete in IDS literature since 2010.

6. Key Network Features for the ML Model

Feature Name	Description	Relevance
Distinct Dst Ports/IP	# unique destination ports per source	Critical core scanning signature
SYN Flag Count	SYN sent without completing handshake	Critical stealth SYN scan
RST Flag Count	RST responses from closed ports	Very High port probe rejection
Flow Duration	Duration of each individual probe connection	High very short in fast scans
Total Fwd Packets	Packets per flow from scanner	High usually 1 per probe
IAT Mean	Mean inter-arrival time between probes	High slow scans have large IAT
ACK Flag Count	ACK packets without prior SYN	Medium ACK scan detection
Unique Dst IPs/Src	# distinct destination hosts contacted	Medium host discovery phase
Bwd Packet Length	Response packet size (RST vs SYN-ACK ratio)	Medium bidirectional profile
TTL Value	Time-to-live in IP header	Low OS fingerprinting
Port Range Entropy	Shannon entropy of destination ports	High randomized vs sequential

Feature Engineering Notes

Port Range Entropy (NEW addition): high entropy signals randomized scans RF-friendly, peer-cited.

Compute all features per (src_ip, 10s window) AND (src_ip, 60s window) independently.

SMOTE applied on training fold only never on validation or test splits prevents leakage.

7. Machine Learning Model Strategy

Component	Detail
Primary Model	Random Forest Classifier high-dimensional tabular data, feature importance
Comparison Model	XGBoost gradient boosting compared on time-windowed aggregated features
Slow Scan Layer	Isolation Forest (unsupervised) on 60-second sliding window per source IP
Imbalance Handling	SMOTE applied on training split only prevents data leakage
Hyperparameter Tuning	GridSearchCV / RandomizedSearchCV on RF and XGBoost
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC-ROC, per-scan-type detection rate
Validation Strategy	Stratified K-Fold (k=10) + temporal holdout (last 20% chronologically)
v1 Output	Binary: BENIGN vs PORT_SCAN with confidence score
v2+ Output	Multi-class: SYN / UDP / ACK / XMAS / Connect Scan

8. Non-Functional Requirements

Requirement	Criterion	Target
Fast Scan Detection	Detection time for aggressive Nmap scan	< 8 seconds
Slow Scan Detection	Detection time for stealthy / slow scan	< 90s (sliding window)
Model F1-Score	On CICIDS2017 PortScan test subset	>= 88%
Model Accuracy	Overall classification on test set	>= 90%
False Positive Rate	Normal connections flagged as scans	< 6%
Dashboard Refresh	Scanner activity heat map update frequency	<= 5 seconds
Blocking Response	Time from detection to iptables DROP rule	< 15 seconds

9. System Architecture Vision

Capture Layer

Scapy BPF filter on LAN interface (SYN-only or full TCP/UDP)
Rolling aggregation per source IP over 10s and 60s windows

Feature Layer

Pandas windowed aggregation -> feature vector per (src_ip, time_window)
Port Range Entropy, SMOTE applied on training split only

Detection Layer

v1: RF binary classifier -> BENIGN / PORT_SCAN + confidence score
v2+: multi-class scan type (SYN / UDP / ACK / XMAS)
Parallel: Isolation Forest on 60s window for slow scan anomalies

Response Layer

Alert: severity score + scan type -> event store
Blocking: iptables DROP via Python subprocess -> audit log
[OPTIONAL] Honeypot: redirect scanner to fake listener -> log all probes

Observability Layer

Real-time dashboard -> Active scanners list, Port activity heat map
Attacker profile (IP, MAC, ports probed, first/last seen timestamps)
Alert feed with confidence, severity, and scan type label

10. Honeypot Response Layer [NEW Hybrid v2.0]

The Honeypot layer is an optional but recommended addition. When the IDS classifies a source IP as PORT_SCAN with confidence ≥ 0.85 , instead of (or in addition to) an iptables DROP, the system redirects the attacker to a lightweight fake listener. This fake service mimics open ports, responds with plausible banners, and logs every probe providing rich attacker intelligence without exposing real services.

10.1 How It Works

- * Detection Layer raises PORT_SCAN alert with confidence ≥ 0.85 for the source IP.
- * Response Layer issues iptables REDIRECT: traffic from attacker IP -> honeypot listener port.
- * Honeypot process (Python socket server) accepts connections, returns fake banners (SSH, HTTP, FTP).
- * All interactions are logged: timestamp, port probed, banner grabbing attempts, payloads sent.
- * After TTL expiry the iptables rule is removed; attacker loses access to the fake listener.

10.2 Implementation Scope

Component	Tool	Effort
Fake listener (Python socket)	Python stdlib socket module	~2 hours
Attacker probe logger	SQLite / structured JSON log	~1 hour
iptables REDIRECT rule	Python subprocess + netfilter	~1 hour
Dashboard integration	Add Honeypot Activity tab	~2 hours
Banner database (SSH/HTTP/FTP)	Static string constants	~30 min

Academic Value of the Honeypot

Adds a Deployment & Response chapter to the CRISP-DM report with concrete attacker intelligence output.

Zero risk to real services completely isolated fake listener process on a dedicated port.

Total implementation time: ~6-7 hours. DEFER to Phase 4 if time is tight.

Do NOT implement IP Polymorphism or Port Hopping sysadmin tasks with zero AI grading value.

11. 5-Person Team Role Assignments [NEW Hybrid v2.0]

Each role maps to a deliverable that contributes to the AI lifecycle documentation grade. No role involves IP Polymorphism or Port Hopping. Every member owns a graded chapter of the final report.

Role 1:	ML Pipeline Lead	Assigned: Adil Lekhbioui (confirmed)
* Download & preprocess CICIDS2017 PortScan and UNSW-NB15 datasets * Implement SMOTE, time-window aggregation, Port Range Entropy feature * Train RF and XGBoost; run GridSearchCV tuning; generate comparison table * Report chapters: Data Preparation + Modeling		
Role 2:	Anomaly & Slow Scan Specialist	Assigned: TBC
* Implement Isolation Forest on 60s sliding window per source IP * Calibrate contamination parameter, validate against labeled slow scans * Integrate IF output with RF confidence in unified alert score * Report chapters: Unsupervised detection layer + evaluation results		
Role 3:	Capture & Feature Pipeline	Assigned: TBC
* Write Scapy listener with BPF filter (SYN-only / full TCP-UDP) * Implement rolling aggregation windows (10s, 60s) per source IP * Compute all features from Section 6 in real time		

Role 4: Dashboard & Alert System

Assigned: TBC

- * Build real-time dashboard (Flask + Socket.IO + D3.js recommended)
- * Implement port heat map, active scanners list, attacker profile view
- * Integrate alert feed (scan type, confidence, severity, timestamp)
- * Implement optional Honeypot Activity dashboard tab
- * Report chapters: Deployment architecture + demo recording

Role 5: Attack Simulation & Documentation

Assigned: TBC

- * Run Nmap scan types (SYN, UDP, ACK, XMAS, Connect) against test VM
- * Implement optional Honeypot listener (Python socket server)
- * Produce per-scan-type detection rate table from live tests
- * Write literature review, ethical considerations, French abstract
- * Report chapters: Problem Statement + Evaluation results

12. Week-by-Week Execution Timeline [NEW Hybrid v2.0]

Each week maps to a concrete CRISP-DM deliverable.

Phase 1 Design & Setup			
Week	Activities	Owner	Deliverable
Wk 1	VM creation (Ubuntu IDS, Kali attacker), network isolation, Git repo, role confirmat	R3+R5	Network diagram, Git repo
Wk 1	Download CICIDS2017 + UNSW-NB15, EDA: class distribution, missing values, st	R1	Data understanding report

Phase 2 Implementation			
Week	Activities	Owner	Deliverable
Wk 2	Data preprocessing: SMOTE, time-window aggregation (10s/60s), Port Range Ent	R1+R3	Clean dataset + feature set
Wk 2	Scapy live capture: BPF filter, rolling aggregation, real-time feature vector output	R3	capture_pipeline.py
Wk 3	Train RF and XGBoost: baseline -> GridSearchCV tuning -> model comparison tat	R1	Trained .pkl + comparison table
Wk 3	Isolation Forest: contamination tuning, integration with RF confidence scores	R2	slow_scan_detector.py
Wk 4	Flask/D3.js dashboard: heat map, active scanners, alert feed, attacker profile	R4	dashboard/ module
Wk 4	iptables blocking engine: Python subprocess, TTL auto-unblock, audit log	R3+R4	response_engine.py

Phase 3 Testing			
Week	Activities	Owner	Deliverable
Wk 5	Attack simulation: Nmap SYN/Connect/UDP/XMAS/ACK scans against test VM	R5	Per-scan detection rate table
Wk 5	False positive testing: legitimate traffic scenarios (file transfers, browsing)	R2+R5	FPR measurement report
Wk 5	Honeypot: fake listener, probe logger, iptables REDIRECT rule [OPTIONAL]	R5	honeypot.py

Phase 4 Validation & Report			
Week	Activities	Owner	Deliverable
Wk 6	K-Fold cross-validation (k=10) + temporal holdout evaluation + AUC-ROC curves	R1+R2	Validation chapter + graphs
Wk 6	End-to-end integration test: live scan -> detection -> alert -> block	AII	Demo recording
Wk 7	Final report: CRISP-DM structure, literature review, ethical considerations	AII	Final report PDF
Wk 7	French abstract (resume), English abstract, architecture diagram, slides	R5	Submission package

13. Project Phases & Deliverables

Phase	Activities	Key Deliverables
1 Design	LAN topology, time-window feature design, scan type taxonomy	Network diagram, feature set document, scan taxonomy
2 Implementation	CICIDS2017 preprocessing, RF+XGBoost+IF training, capture pipeline, r	Trained .pkl models, IDS pipeline, port heat map dashboard
3 Testing	Simulate Nmap SYN/Connect/UDP/XMAS; measure per-type detection r	Per-scan detection table, ROC curves, FPR results
4 Validation	K-Fold cross-validation, temporal holdout, end-to-end scenario	Final report, demo recording, multi-model comparison table

14. Technical Stack Summary

Component	Tool / Library	Notes
OS / Virtualization	Ubuntu Server (IDS), Kali Linux (attacker), GNS3 / VirtualBox	
Programming Language	Python 3.10+	
Traffic Capture	Scapy (BPF), PyShark, sliding window aggregation	

ML Libraries	Scikit-learn, XGBoost, imbalanced-learn, Pandas, NumPy	
Web Dashboard	Flask + Socket.IO + D3.js	Alt: Grafana + InfluxDB
Blocking Engine	Linux iptables DROP/REDIRECT via Python subprocess, TTL unblock	
Attack Simulation	Nmap (SYN/ACK/UDP/XMAS/Connect), Masscan, Scapy scripts	
Honeypot (optional)	Python socket + SQLite probe log + iptables REDIRECT	NEW v2.0
Version Control	Git + GitHub feature branches per module	

15. Expected Outcomes & Success Criteria

- [OK]

Model achieves F1-Score >= 88% on CICIDS2017 PortScan test subset.
- [OK]

Nmap SYN stealth scan detected and alert triggered within 8 seconds of initiation.
- [OK]

Slow-rate scan (1 probe/second) detected within the 90-second analysis window.
- [OK]

Dashboard heat map correctly highlights targeted ports and the scanning source IP.
- [OK]

Attacker profile correctly identifies scan type (SYN / UDP / ACK / XMAS) in most cases.
- [OK]

Legitimate traffic produces a false positive rate below 6%.
- [OK]

[OPTIONAL] Honeypot successfully logs attacker probes after iptables REDIRECT is applied.

CRISP-DM Chapter Coverage Checklist	
[] 1. Business Understanding	Sections 1-3 of this BRD
[] 2. Data Understanding	Section 5 (datasets) + Section 6 (features)
[] 3. Data Preparation	SMOTE, time-window aggregation (Section 7)
[] 4. Modeling	RF + XGBoost + Isolation Forest (Section 7)
[] 5. Evaluation	K-Fold, temporal holdout, comparison table (Section 8)
[] 6. Deployment	Architecture (Sec 9), Honeypot (Sec 10), Dashboard
[] Extras (for top grade)	Literature review, ethical considerations, French abstract